

EX NO:5

Design of High Pass Filter using Windowing Techniques

5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

Program (Matlab Code):

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000;    % High-pass cutoff frequency (Hz)

fs = 8000;    % Sampling frequency (Hz)

N = 50;       % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- 1. FIR High Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'high', rectwin(N+1));

disp('--- High-Pass Coefficients (Rectangular Window) ---');

disp(b_rect);

%% --- 2. FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');

disp(b_hamm);

%% --- 3. FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');

disp(b_hann);

%% --- Frequency Response Plots ---
```

```

figure;

freqz(b_rect,1);

title('High-pass FIR using Rectangular Window');

figure;

freqz(b_hamm,1);

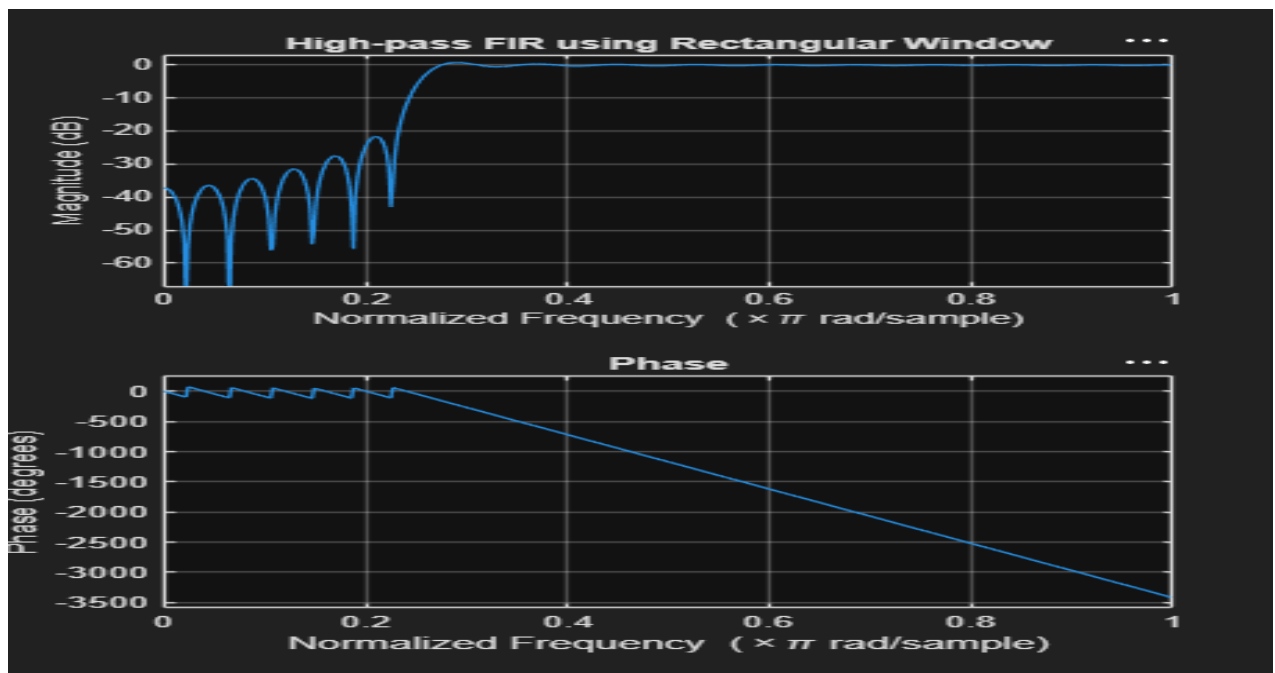
title('High-pass FIR using Hamming Window');

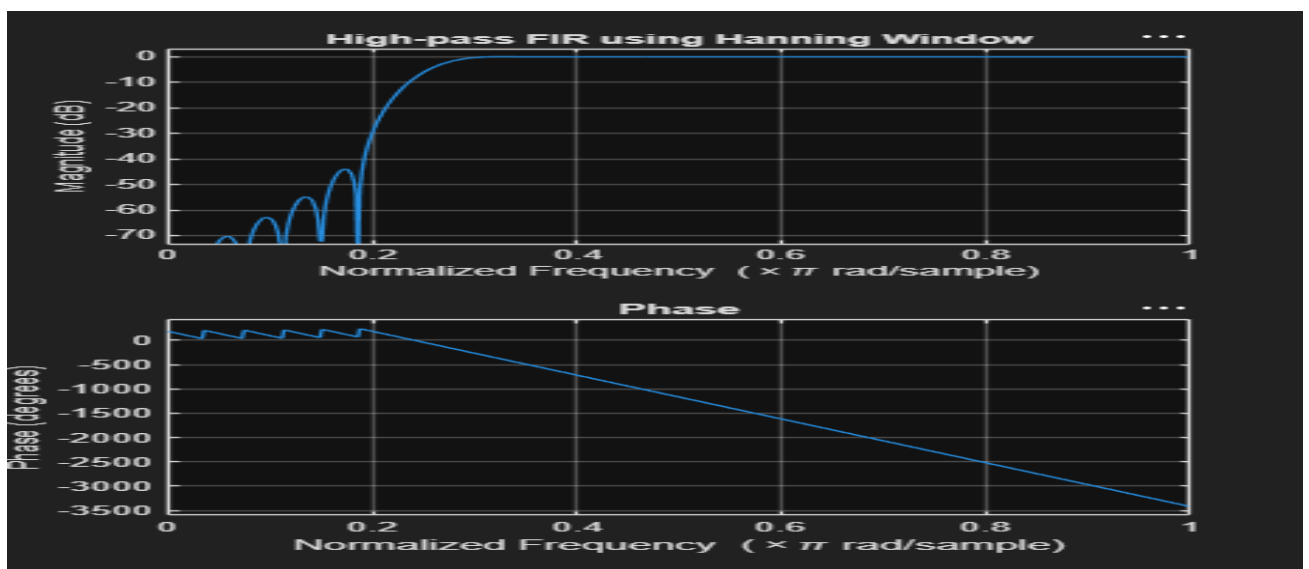
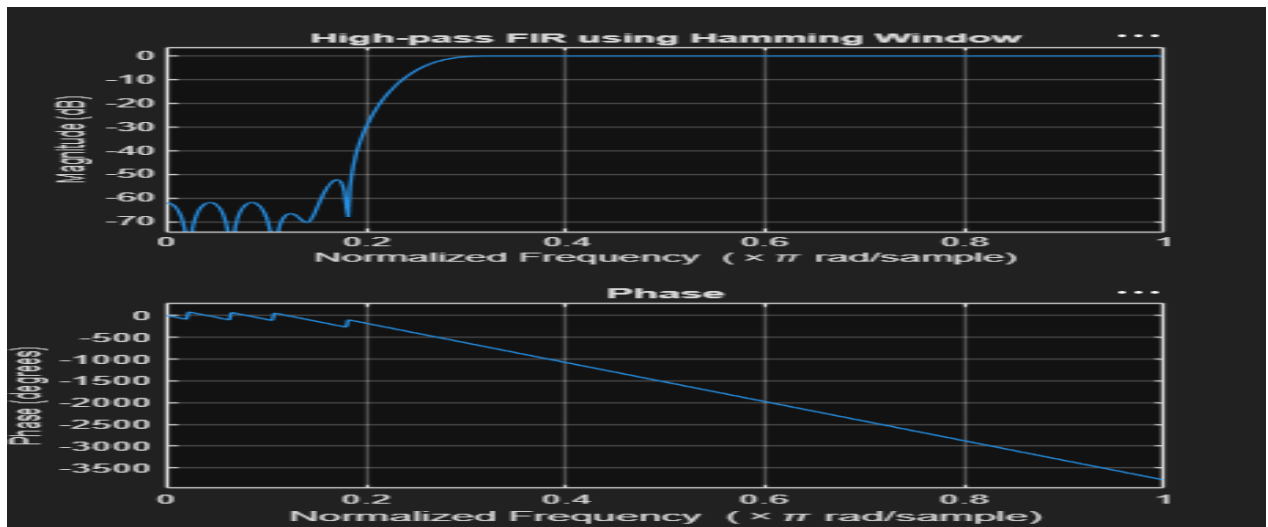
figure;

freqz(b_hann,1);

title('High-pass FIR using Hanning Window');

```





High-Pass Coefficients (Rectangular Window) ---

```

-0.0089
  0
  0.0097
  0.0143
  0.0106
  0
-0.0117
-0.0175
-0.0131
  0
  0.0148
  0.0225
  0.0171
  0
-0.0202
-0.0314
-0.0247
  0
  0.0318
  0.0524

```

0.0445
0
-0.0741
-0.1572
-0.2223
0.7407
-0.2223
-0.1572
-0.0741
0
0.0445
0.0524
0.0318
0
-0.0247
-0.0314
-0.0202
0
0.0171
0.0225
0.0148
0
-0.0131
-0.0175
-0.0117
0
0.0106
0.0143
0.0097
0
-0.0089

--- High-Pass Coefficients (Hamming Window) ---
-0.0007

5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

Program (Matlab Code):

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000;    % High-pass cutoff frequency (Hz)
fs = 8000;    % Sampling frequency (Hz)
N = 50;      % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');

disp(b_hamm);

%% --- Frequency Response Plots ---

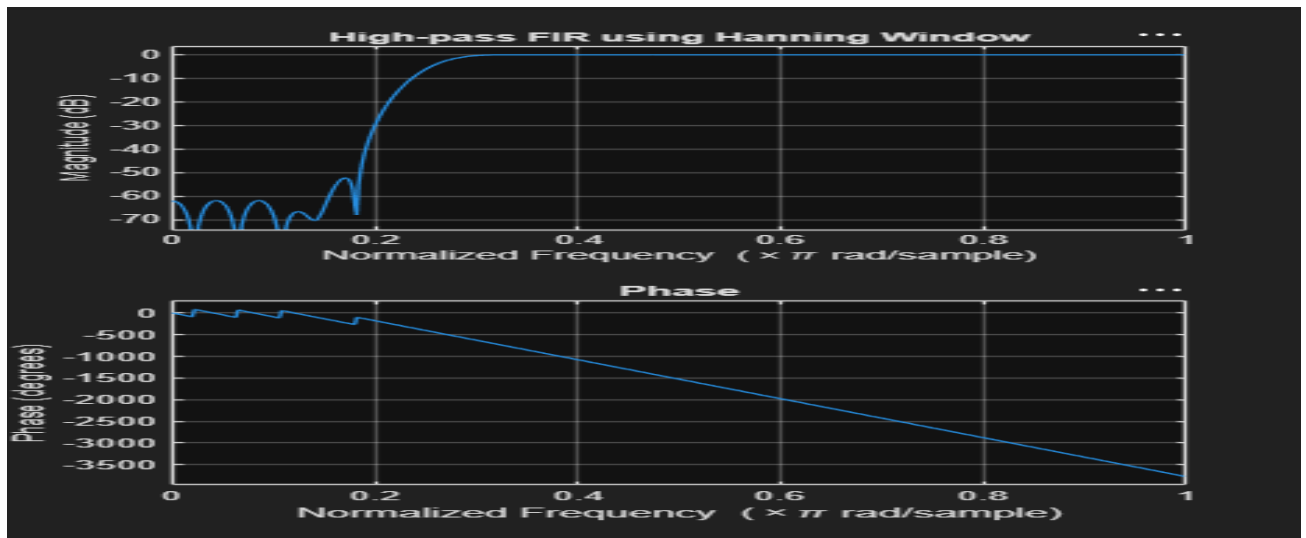
figure;

freqz(b_hamm,1);

title('High-pass FIR using Hamming Window');

title('High-pass FIR using Hanning Window');
```

Output:



High-Pass Coefficients (Hamming Window) ---

```

-0.0007
  0
  0.0009
  0.0016
  0.0015
  0
-0.0024
-0.0044
-0.0039
  0
  0.0060
  0.0103
  0.0088
  0
-0.0128
-0.0217
-0.0184
  0
  0.0268
  0.0464
  0.0410
  0
-0.0725
-0.1567
-0.2240
  0.7493
-0.2240
-0.1567
-0.0725
  0
  0.0410
  0.0464
  0.0268
  0

```

-0.0184
-0.0217
-0.0128
0
0.0088
0.0103
0.0060
0
-0.0039
-0.0044
-0.0024
0
0.0015
0.0016
0.0009
0
-0.0007

5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing techniques

Program (Matlab Code):

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000;    % High-pass cutoff frequency (Hz)
fs = 8000;    % Sampling frequency (Hz)
N = 50;      % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hanning Window ---

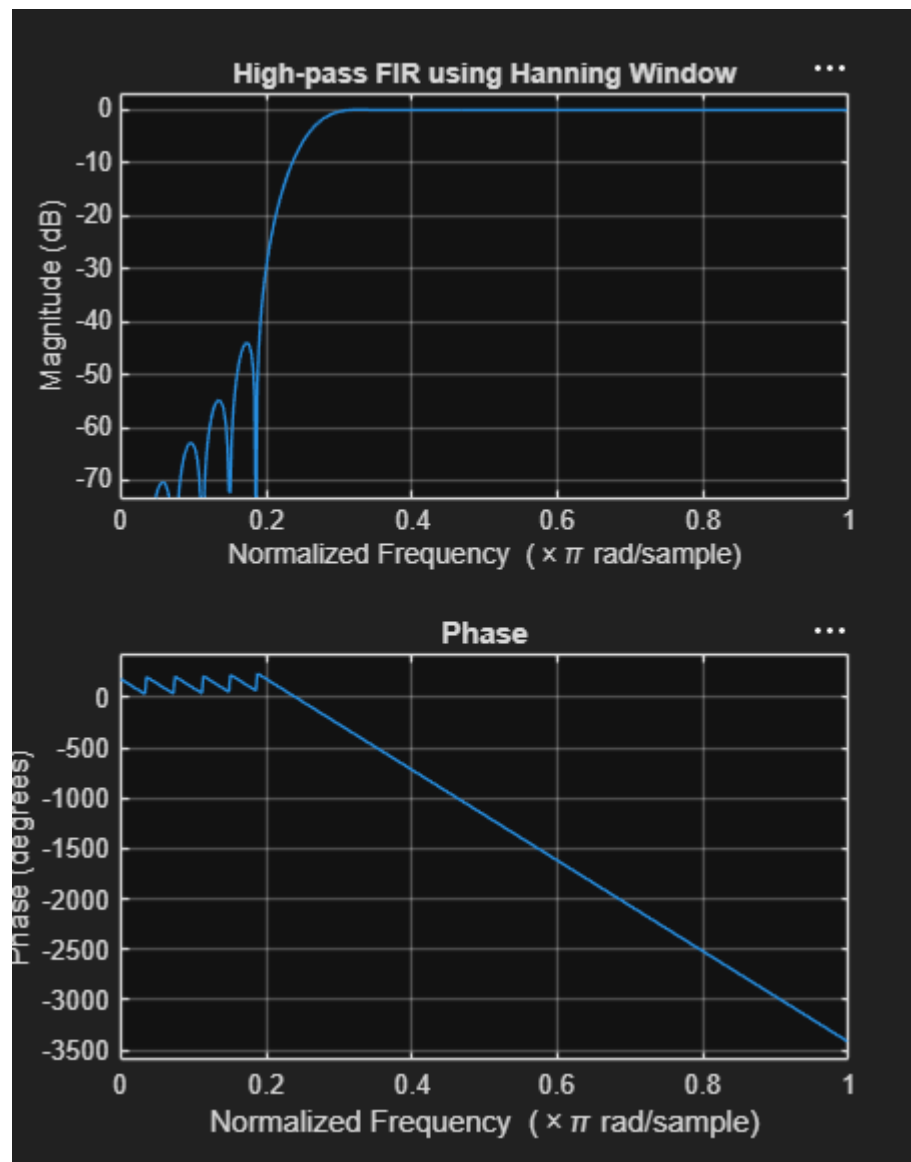
b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plots ---

figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```


Output:



High-Pass Coefficients (Hanning Window) ---

-0.0000
0
0.0003
0.0008
0.0009
0
-0.0020
-0.0038
-0.0035
0
0.0057
0.0100
0.0087
0
-0.0127

-0.0216
-0.0183
0
0.0267
0.0464
0.0410
0
-0.0726
-0.1568
-0.2243
0.7500
-0.2243
-0.1568
-0.0726
0

0.0410
0.0464
0.0267
0
-0.0183
-0.0216
-0.0127
0
0.0087
0.0100
0.0057
0
-0.0035
-0.0038
-0.0020
0
0.0009
0.0008
0.0003
0
-0.0000